

Jump Shinobi

Final Game Prototype Report

Deliverable 3 – Showcase Submission

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This report summarizes the final design, implementation, aesthetic direction, playtesting outcomes, technical improvements, and presentation focus of Jump Shinobi.

Project Overview

Jump Shinobi is a 2D action-platformer centered on precision traversal, responsive controls, and player mastery. The game combines wall-jumping, dashing, melee combat, environmental guidance, and stylized world-building to create a focused experience that rewards timing and movement skill.

The objective of the project was not only to produce a functional final prototype, but also to demonstrate how the design evolved through iteration. This includes improvements to difficulty pacing, player readability, guidance, visual consistency, and overall game feel.

Area	Final State
Core Mechanics	Movement, jump, wall-jump, dash, and melee combat are integrated into moment-to-moment gameplay.
Level Structure	Tutorial / lore opening area followed by progressively more demanding traversal-focused levels.
Aesthetic Direction	Stylized shinobi-inspired presentation with environmental contrast and atmospheric cues.
Narrative Delivery	Environmental storytelling through props, room design, signs, and lore-based spaces.
Iteration Focus	Fairness, clarity, responsiveness, and stronger player guidance based on testing feedback.

1. Introduction

Jump Shinobi was designed as a movement-first platformer where the satisfaction of play comes from precision, rhythm, and steady skill growth. Rather than relying solely on combat or narrative exposition, the game places movement at the center of the user experience. Traversal itself becomes the main challenge and the main source of fun, which aligns strongly with platformer design principles that reward practice, timing, and pattern recognition.

The initial concept focused on creating a shinobi-themed game that felt agile and deliberate. This theme influenced not only the mechanics, but also the level design and visual presentation. The player is expected to move quickly, adapt to obstacles, and learn through interaction. As a result, the final prototype aims to create an experience where the player feels increasingly capable as they progress through each section.

From a design perspective, the game reflects core course ideas around flow and engagement. A successful level should neither be trivial nor overwhelming. Instead, each challenge should sit just above the player's current comfort level, encouraging repeated attempts and creating a sense of accomplishment once mastered. The final version of Jump Shinobi was built with this principle in mind, leading to several revisions in challenge pacing, movement tuning, and environmental readability.

2. Core Gameplay and Player Experience

The core gameplay loop of Jump Shinobi is simple but intentionally tight: observe the space, read the intended path, execute movement cleanly, and continue forward. This basic loop is enhanced by layered mechanics. Standard movement gives the player foundational control, while jumping, wall-jumping, and dashing expand the number of possible solutions to traversal problems. Melee combat adds short bursts of interaction that break up movement sections without shifting the game away from its platforming identity.

A key design goal was responsiveness. In movement-heavy games, player frustration often comes less from difficulty itself and more from controls that feel unreliable. For that reason, significant attention was given to how the character accelerates, changes direction, jumps, and transitions off walls. The result is a control scheme intended to feel immediate and predictable so that when players fail, they understand the failure as a gameplay mistake rather than a systems issue.

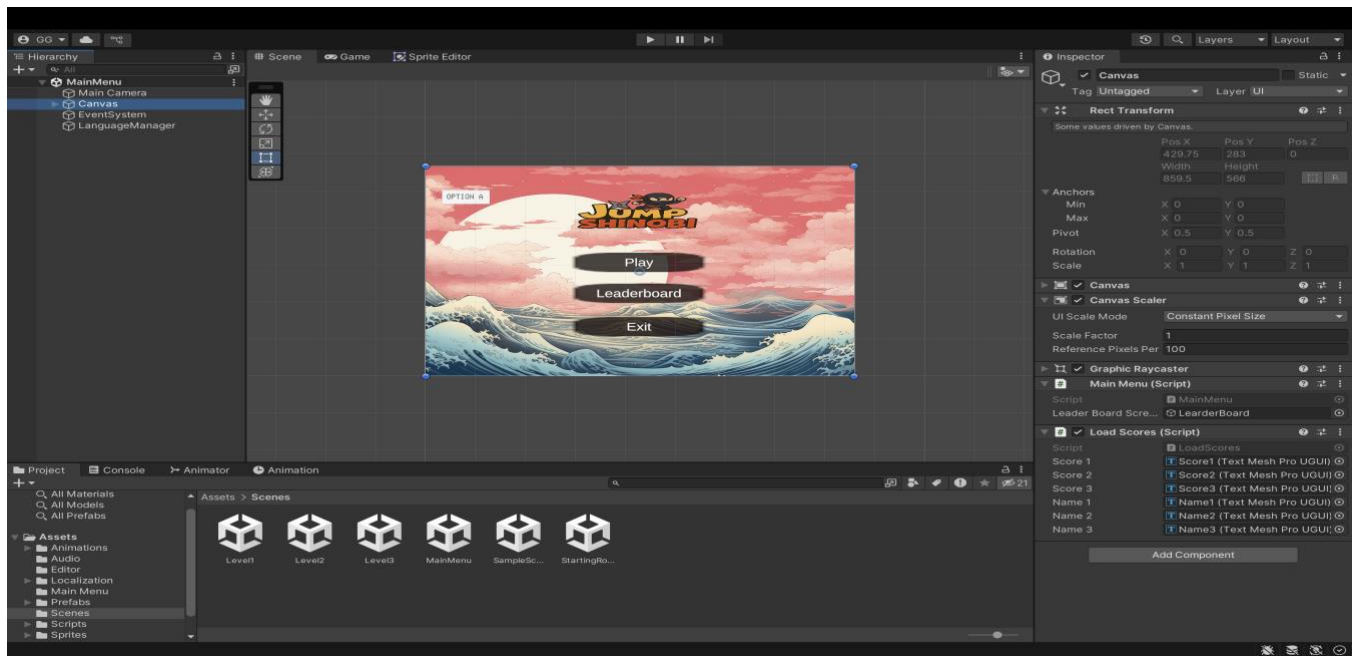
The overall player experience is meant to move from curiosity to competence. Early interactions introduce the world and establish the game's movement language. Later sections ask the player to combine previously learned mechanics in more demanding ways. This creates a progression where improvement feels earned, and where success is tied to mastery rather than randomness.

3. Art, Aesthetics, and Sound Design

The visual direction of Jump Shinobi supports both atmosphere and usability. The shinobi theme encouraged an aesthetic built around darker spaces, ancient structures, moody lighting, and selective highlights. Instead of filling the screen with unnecessary visual noise, the environment uses contrast to communicate importance. Brighter objects, lit pathways, and distinct silhouettes help direct the player's attention toward meaningful spaces and interactive elements.

This visual approach also supports player guidance. In platformers, players are constantly making fast judgments about where they can go, what is safe, and what looks important. By emphasizing certain environmental features through lighting and composition, the game can guide players naturally without relying too heavily on intrusive arrows or tutorial text. This preserves immersion while still improving clarity.

The user interface was kept minimal so that the game world remains the primary focus. Any on-screen information is intended to support play rather than dominate it. This is especially important in a precision platformer, where screen clutter can distract from the timing and awareness required to move cleanly through a space.



Visual Design of the Main Menu and GUI for the Game

Sound design contributes to the same goals. Even if some audio elements remain partial or placeholder, their purpose is clear: reinforce actions, provide feedback, and strengthen atmosphere. Audio cues for movement, attacks, danger, or interaction help players interpret the outcome of their inputs. They also make successful actions feel more impactful, which improves game feel and overall immersion.

4. Narrative and Systems Integration

Jump Shinobi does not rely on lengthy cutscenes or heavy exposition. Instead, its narrative is communicated through environmental storytelling, room structure, and the relationship between the player's abilities and the world they move through. This style of delivery fits the project well because it allows the player to remain engaged in gameplay while still building a sense of place and identity.

The opening lore-oriented area serves an important role in this structure. It acts as more than just a starting room: it establishes tone, introduces the world, and gives context to the player's journey. Objects such as shrines, statues, or signs can be used to imply history, danger, and purpose without stopping the pace of the game.

The mechanics themselves also reinforce the narrative identity. A shinobi fantasy suggests agility, precision, discipline, and adaptability. Those same qualities are reflected in the player's toolkit. Wall-jumping, fast movement, and carefully timed traversal all support the idea that the player character is highly mobile and skilled. Because of this, the narrative theme is not separate from the gameplay systems; it is embedded within them.

This integration helps the final prototype feel more cohesive. When the art style, movement systems, and level spaces all support the same fantasy, the result is a stronger player experience than if these elements existed independently.

5. Level Design, Progression, and Balance

Level design in Jump Shinobi was developed around the principle of gradual mastery. The game does not begin by immediately demanding high-level execution. Instead, earlier spaces establish the rhythm of movement and the expectations of the environment. This gives players room to experiment with inputs and understand the relationship between the character's abilities and the layout of the level.

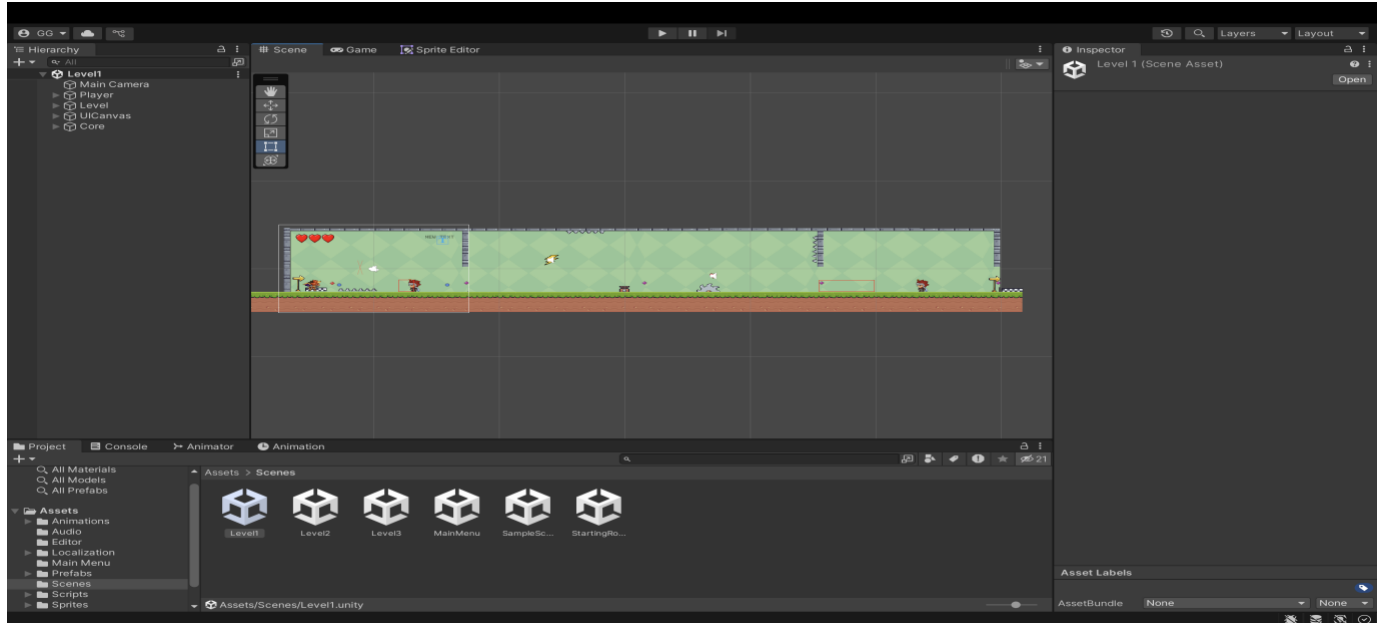
As the game progresses, the complexity of challenges increases. This can happen through longer jump chains, more demanding platform placement, tighter timing windows, or combinations of mechanics in a single sequence. The key balancing challenge was ensuring that each new section felt meaningfully harder without becoming discouraging. Difficulty should create tension and engagement, not confusion or helplessness.

To maintain this balance, several aspects of level structure were revised over time. Platform distances, hazard placement, and recovery opportunities all affect how fair a level feels. If spacing is too strict, players may feel punished before they have enough information to improve. If spacing is too generous, the challenge loses intensity. The final prototype attempts to sit between these extremes by creating sections that are demanding but readable.

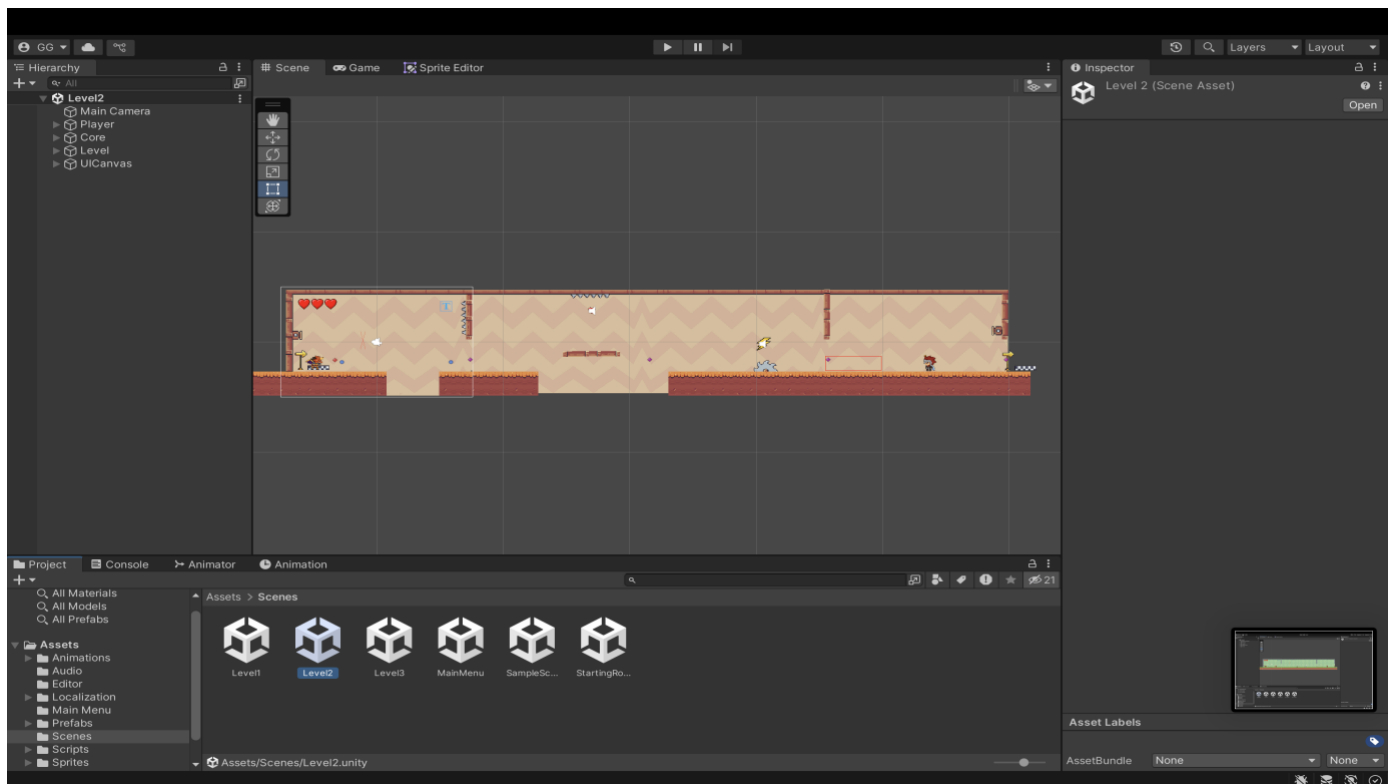
Progression is also shaped by player psychology. A platforming challenge feels more rewarding when the player can recognize why they failed and how to improve on the next attempt. This is why clear level readability is so important. Players should be able to identify a route, anticipate danger, and understand which mechanics a section expects them to use.

Where applicable, checkpoints or natural pauses in intensity help preserve motivation. These moments break up tension and allow the player to feel forward momentum, which is important in maintaining flow. The balance between risk and reward therefore appears not only in difficulty itself,

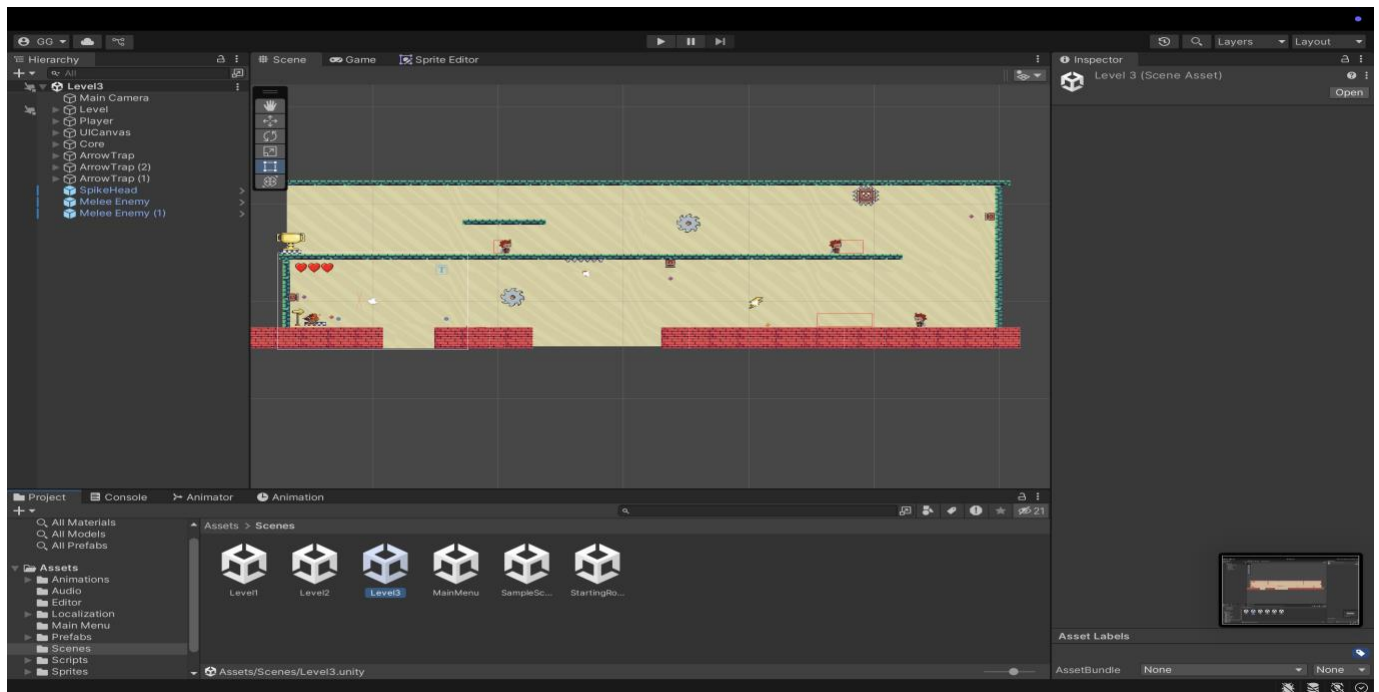
but also in pacing and recovery.



Screenshot of Level 1 of the Game



Screenshot of Level 3 of the Game



Screenshot of Level 3 of the Game

6. Assets & Visual Development

The development of visual assets played an important role in shaping both the identity and usability of the game. Core assets, including the game logo and character sprite sheets, were designed using Photoshop to establish a consistent visual style aligned with the shinobi theme. These assets were not only aesthetic choices but also functional components that contributed to player readability and engagement.

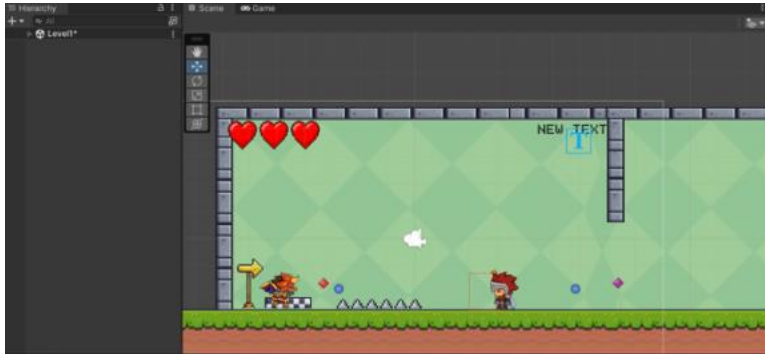
Design of the game logo (created using Photoshop):



Drawn Character model (using Photoshop):



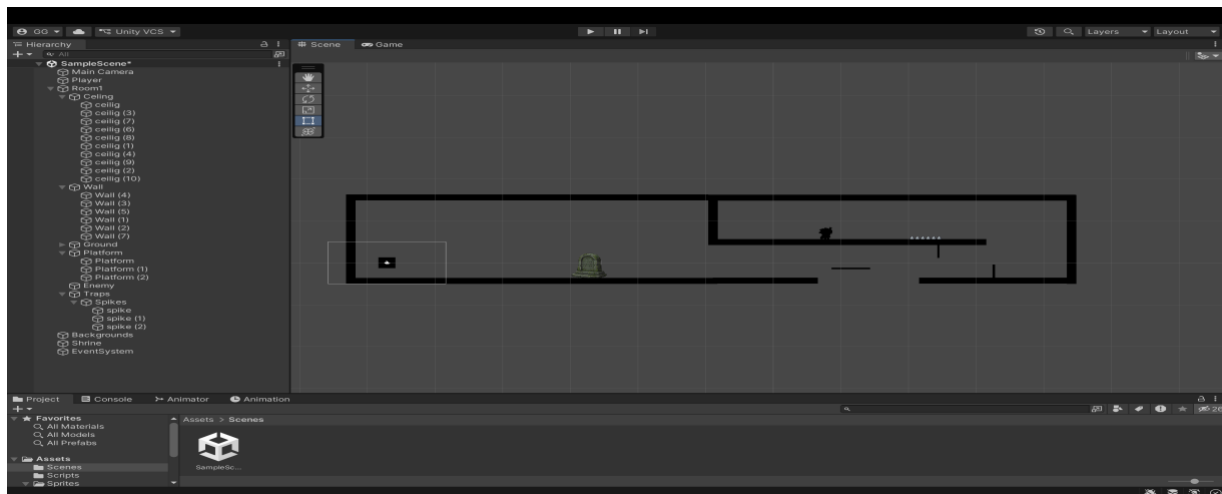
Initial Design (using Unity) :



In addition to finalized assets, placeholder visuals were used during early development stages to rapidly prototype mechanics and test gameplay flow. This allowed for quick iteration without being constrained by finalized art. As development progressed, these placeholders were gradually replaced with more refined assets, ensuring that visual clarity, thematic consistency, and gameplay readability improved alongside the mechanics. This iterative approach helped balance development efficiency with visual polish in the final prototype.

6. Playtesting and Iterative Refinement

Playtesting was one of the most important parts of the project because it revealed the difference between what the game designer intended and what players actually experienced. Many design issues do not become obvious until another person attempts to play the game without prior explanation. Observing those moments helped identify where the prototype was unclear, overly punishing, or mechanically inconsistent.



Initial Greybox Used for Prototyping and testing

One recurring issue involved platforming difficulty. Certain areas demanded too much precision too early, which made the experience feel abrupt rather than skill-building. In response, platform spacing and timing expectations were adjusted so that difficulty could escalate more naturally. This did not mean removing challenge entirely; instead, it meant making the challenge feel fairer and easier to learn from.

Another issue involved navigation and player guidance. Some players hesitated because they were unsure where to move next or which surfaces were intended to be used. To address this, environmental composition and visual cues were strengthened. Lighting, object placement, and the layout of traversal routes were refined to make intended paths more legible while still preserving a natural presentation.

Control feel was also improved through iteration. Because movement is the core of the game, even small frustrations in jumping or wall interaction can have a large negative effect. Repeated testing helped refine timing, responsiveness, and the consistency of transitions between actions. These improvements strengthened player trust in the game systems, which is essential in a skill-based platformer.

The most valuable outcome of playtesting was not simply a list of isolated fixes, but a better understanding of how pacing, clarity, and feedback interact. A player is more willing to accept difficulty when the game communicates clearly, responds reliably, and makes improvement feel possible. That principle shaped many of the final revisions in the prototype.

7. Technical Improvements and Stability

Technical refinement supported the design goals of the final prototype. A game may have strong ideas, but if core systems behave inconsistently, the player experience suffers immediately. For that reason, stability and responsiveness were treated as important parts of polish rather than as secondary concerns.

Movement-related bugs were particularly important to resolve. Collision issues, wall interactions, unexpected momentum behavior, or timing inconsistencies can undermine confidence in the controls. Addressing these problems helped ensure that movement challenges tested player skill rather than exposing technical roughness.

In addition to gameplay fixes, the project benefited from more general optimization and cleanup. Reducing unnecessary logic, organizing assets more effectively, and reviewing scene performance all contribute to smoother play. Even when improvements are not dramatic enough for players to identify individually, they affect the overall feel of the game by making the experience more stable and consistent.

Debugging also contributed to the fairness of the game. A difficult platforming sequence is acceptable when it behaves consistently. It becomes frustrating when failure appears to come from unpredictable or inconsistent system behavior. By resolving technical issues, the final prototype became more reliable and therefore more enjoyable to learn and play.

8. Reflection on Design Goals and Final State

The final version of Jump Shinobi reflects a strong connection between concept and implementation. The project began with the goal of building a shinobi-themed platformer centered on movement mastery, and the finished prototype maintains that identity throughout its systems, level structure, and presentation. The result is not just a collection of features, but a more cohesive experience shaped by repeated refinement.

One of the biggest lessons from the project was the importance of alignment between difficulty and readability. A challenging game can still feel welcoming when the player understands what the game expects. This reinforced the value of subtle guidance, strong feedback, and consistent controls.

Another major takeaway was the role of iteration in transforming a promising idea into a stronger final product. Many of the project's most meaningful improvements did not come from adding entirely new systems; they came from tuning, rebalancing, clarifying, and polishing what was already there. This process highlighted how much quality in game design comes from refinement rather than raw feature count.

Overall, Jump Shinobi demonstrates how course concepts such as player engagement, flow, progression, and iterative testing can directly influence a final game prototype. The project serves as both a functional game and a practical application of the design principles explored throughout the course.

9. Conclusion

Jump Shinobi concludes as a polished final prototype that brings together movement mechanics, level design, atmospheric presentation, and iterative problem-solving into a unified experience. The game's strengths come from its focused identity: it understands that its core appeal lies in traversal, responsiveness, and the satisfaction of gradual mastery.

Across development, the project evolved through repeated feedback and revision. Adjustments to balance, clarity, controls, and presentation helped transform earlier concepts into a more complete and better-paced final build. The final result demonstrates not only technical implementation, but also a thoughtful design process shaped by testing and reflection.

In that sense, the project represents the culmination of the course material. It applies theoretical ideas about challenge, flow, player motivation, and iteration in a concrete, playable form. More importantly, it reflects a growing understanding that strong game design depends on how mechanics, presentation, and player experience work together as one system.